Quantum Fluctuations Around Rope Knots

The Fluctuation Spectrum and the Electron Log-Determinant Requirement

Mark Palmer · with computational collaboration by Claude (Anthropic)

Charlotte, NC · palmer100@gmail.com

Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

Abstract

Around a classical rope knot, small fluctuations have a discrete spectrum whose determinant controls the knot's quantum weight. This paper reports the computed fluctuation spectrum for a Pöschl–Teller-type potential (bound-state energies [−4.000, −1.001] in natural units) and the log-determinant that a knot must supply to match the electron's scale, computed as 108.12. We are explicit that the latter is a REQUIREMENT the spectrum must meet, computed as a target, not a derivation of the electron. All numbers in this paper are outputs of the rope_solver computation in this package (reproducible via benchmarks/reproduce_results.py), not inserted constants.

Scope of this paper

CLAIM: the fluctuation spectrum around a rope knot is computed (bound states −4.000, −1.001), and the log-determinant required to match the electron scale is 108.12.

NOT CLAIMED: that a rope knot HAS this determinant — 108.12 is the target the spectrum would need to hit. Whether a knot meets it is Open. This is a computed requirement, not a derivation of the electron mass.

1. Fluctuations and the Quantum Weight

A classical soliton's contribution to quantum amplitudes is weighted by the determinant of the operator governing small fluctuations around it — the product of the fluctuation frequencies. A solvable model of the transverse potential is the Pöschl–Teller well, whose bound-state spectrum the solver returns as [−4.000, −1.001] in natural units.

2. The Electron Requirement, as a Target

Matching the electron's scale imposes a specific value on the fluctuation log-determinant. The solver computes that required value as 108.12. This is stated honestly as a REQUIREMENT: it is the number a rope knot's fluctuation spectrum would have to produce to reproduce the electron. Whether any actual knot spectrum yields 108.12 is not established here and is listed as open.

Status

Pöschl–Teller bound-state spectrum [−4.000, −1.001] — Derived (solver).

Required electron log-det = 108.12 — Modeled (computed as a TARGET).

A rope knot actually yielding 108.12 — Open, not claimed.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.